

Institute of Actuaries of India

Subject CT3 – Probability & Mathematical Statistics

October 2014 Examinations
Indicative Solutions

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other approaches leading to a valid answer and examiners have given credit for any alternative approach or interpretation which they consider to be reasonable

- Q. 1)**
- i) **Dichotomous data:** Data that are classified into one of two mutually exclusive values.
E.g.: 'yes' and 'no'.
 - ii) **Nominal data:** A set of data is said to be nominal if the values/observations belonging to it can be assigned a code in the form of a number where the numbers are simply labels. One can count but not order or measure nominal data.
E.g.: In insurance policy data, males could be coded as 0 and females as 1.
 - iii) **Ordinal data:** In statistics, ordinal data is a statistical data type consisting of numerical scores that exist on an ordinal scale, i.e., an arbitrary numerical scale where the exact numerical quantity of a particular value has no significance beyond its ability to establish a ranking over a set of data points.
E.g.: Questionnaire responses such as “strongly in favour / ... / strongly against”.

[3]

- Q. 2)** Given 40% of the pens are red while the rest are black, probability of picking one red pen in any attempt is 0.4.

- i) Consider X as the random variable representing the number of pens examined till four red pens are found (with finding a red pen regarded as a success with probability 0.4).

Then, $X \sim$ (Type 1) Negative Binomial distribution with parameters $k = 4$ and $\theta = 0.4$.

$$\text{Reqd. probability} = \binom{10-1}{4-1} (0.4)^4 (1-0.4)^{10-4} = \frac{9!}{3!6!} (0.4)^4 (0.6)^6 = 0.1003$$

- ii) Here, consider X as the random variable representing the number of pens examined till two red pens are found (with finding a red pen regarded as a success with probability 0.4).

Then, $X \sim$ (Type 1) Negative Binomial distribution with parameters $k = 2$ and $\theta = 0.4$.

$$\text{Expected number of pens} = E(X) = \mu = k/\theta = 2/0.4 = 5.$$

[4]

- Q. 3)** Let X_i denote the number option chosen by the participant for the i^{th} question, for $i = 1, 2 \dots 200$. Then, X_i are i.i.d. discrete uniform random variables on 1, 2 and 3 with probability of picking any option is $\frac{1}{3}$.

$$E(X_i) = \frac{1}{3} (1 + 2 + 3) = 2$$

$$\text{Var}(X_i) = \frac{1}{3} (1^2 + 2^2 + 3^2) - 2^2 = \frac{2}{3}$$

$$\text{Now, } T = \sum_{i=1}^{200} X_i$$

Given X_i are independent,

$$E(T) = E\left(\sum_{i=1}^{200} X_i\right) = \sum_{i=1}^{200} E(X_i) = 200 * 2 = 400$$

$$Var(T) = Var\left(\sum_{i=1}^{200} X_i\right) = \sum_{i=1}^{200} Var(X_i) = 200 * \frac{2}{3} = \frac{400}{3}$$

Using Central Limit Theorem, $T \sim N\left(400, \frac{400}{3}\right)$

Thus,

Required Probability

$$= P(380 \leq T \leq 420)$$

$$= P(379.5 < T < 420.5) \quad \dots \text{applying continuity correction}$$

$$= P\left(\frac{379.5 - 400}{\sqrt{400/3}} < \frac{T - 400}{\sqrt{400/3}} < \frac{420.5 - 400}{\sqrt{400/3}}\right)$$

$$= P(-1.78 < Z < 1.78) \quad \dots Z \sim N(0, 1)$$

$$= 2 * \Phi(1.78) - 1$$

$$= 2 * 0.96246 - 1$$

$$= 0.92492$$

[5]

Q. 4) i) The following table contains the cumulative frequency:

Marks	Frequency	Cum Freq.
55	1	1
60	3	4
63	7	11
67	8	19
70	5	24
72	4	28
74	11	39
75	7	46
81	9	55
85	8	63
89	5	68
91	2	70
97	2	72

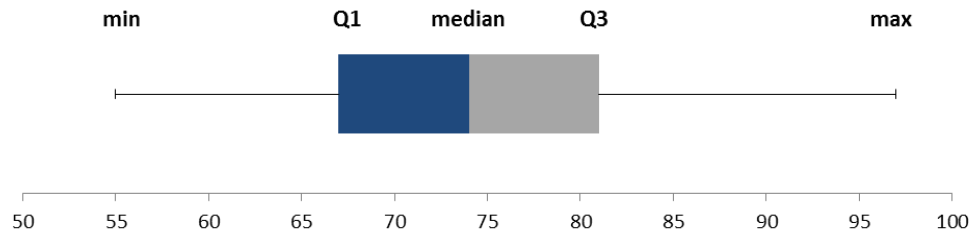
The appropriate quartiles are obtained with reference to the cumulative frequency above:

Statistic	n th Obs.	Marks
min	1	55
Q1	18.5	67
median	36.5	74
Q3	54.5	81
max	72	97

Here:

- $Q1 = (72 + 2)/4$ i.e. 18.5th observation
- Median = $(2 * 72 + 2)/4$ i.e. 36.5th observation
- $Q3 = (3 * 72 + 2)/4$ i.e. 54.5th observation

The boxplot is drawn as below:



- ii) The inter-quartile range = $Q3 - Q1 = 81 - 67 = 14$.

[4]

Q. 5) i) $M'_Z(t) = -0.18e^{-2t} - 0.24e^{-t} + 0.24e^t + 0.18e^{2t}$
 $M''_Z(t) = 0.36e^{-2t} + 0.24e^{-t} + 0.24e^t + 0.36e^{2t}$

Thus:

$$E(Z) = M'_Z(0) = -0.18 - 0.24 + 0.24 + 0.18 = 0$$

$$Var(Z) = E(Z^2) - [E(Z)]^2 = M''_Z(0) - 0 = 0.36 + 0.24 + 0.24 + 0.36 = 1.20$$

- ii) Substituting e^t with t in the MGF, we get the probability generating function (PGF) of Z as: $G_Z(t) = 0.09t^{-2} + 0.24t^{-1} + 0.34t^0 + 0.24t^1 + 0.09t^2$, $-\infty < t < \infty$

This means Z is a discrete random variable which takes values -2, -1, 0, 1 & 2 with probabilities 0.09, 0.24, 0.34, 0.24 & 0.09 respectively. Therefore, considering that X and Y are identically distributed, the permissible values that X (or Y) take will be -1, 0 & 1.

Since $E(Z) = 0$ and given X and Y are identically distributed, $E(X) = E(Y) = 0$.

This means: $-1 * P(X = -1) + 0 * P(X = 0) + 1 * P(X = 1) = 0$

$$\Rightarrow P(X = -1) = P(X = 1).$$

Given X and Y are identically distributed, $Var(X) = Var(Y)$ and since they are uncorrelated, $Cov(X, Y) = 0$.

Thus, $Var(Z) = 1.2$ implies $Var(X) = Var(Y) = 0.6$.

This means: $(-1)^2 * P(X = -1) + 0^2 * P(X = 0) + 1^2 * P(X = 1) = 0.6$

$$\Rightarrow P(X = -1) + P(X = 1) = 0.6.$$

Therefore, $P(X = -1) = P(X = 1) = 0.3$.

& $P(X = 0) = 1 - P(X = -1) - P(X = 1) = 1 - 0.3 - 0.3 = 0.4$.

Thus, X (or Y) takes values -1, 0 & 1 with probabilities 0.3, 0.4 & 0.3.

- iii) Define $p_{i,j} = P(X = i, Y = j)$ where $i, j = -1, 0, 1$.
As X and Y are identically distributed, $p_{i,j} = p_{j,i} \forall i, j$.

$$\begin{aligned}
 0.24 &= P(Z = 1) \\
 &= P(X + Y = 1) \\
 &= P(X = 1, Y = 0) + P(X = 0, Y = 1) \\
 &= p_{1,0} + p_{0,1} \\
 &= 2 * p_{1,0}
 \end{aligned}$$

Thus, $p_{1,0} = p_{0,1} = 0.12$.

Similarly,

$$0.24 = P(Z = -1) = p_{-1,0} + p_{0,-1} = 2 * p_{-1,0}$$

Thus, $p_{-1,0} = p_{0,-1} = 0.12$.

Again,

$$0.09 = P(Z = 2) = P(X + Y = 2) = P(X = 1, Y = 1) = p_{1,1}$$

$$0.09 = P(Z = -2) = P(X + Y = -2) = P(X = -1, Y = -1) = p_{-1,-1}$$

Now, (using part ii)

$$\begin{aligned}
 0.4 &= P(X = 0) = \sum_{j=-1,0,1} P(X = 0, Y = j) = p_{0,-1} + p_{0,0} + p_{0,1} = 0.12 + p_{0,0} + 0.12 \\
 &\Rightarrow p_{0,0} = 0.16
 \end{aligned}$$

$$\begin{aligned}
 0.3 &= P(X = 1) = \sum_{j=-1,0,1} P(X = 1, Y = j) = p_{1,-1} + p_{1,0} + p_{1,1} = p_{1,-1} + 0.12 + 0.09 \\
 &\Rightarrow p_{1,-1} = 0.09
 \end{aligned}$$

$$\begin{aligned}
 0.3 &= P(X = -1) = \sum_{j=-1,0,1} P(X = -1, Y = j) = p_{-1,-1} + p_{-1,0} + p_{-1,1} \\
 &= 0.09 + 0.12 + p_{-1,1} \\
 &\Rightarrow p_{-1,1} = 0.09
 \end{aligned}$$

Thus, the joint probability of X and Y is given as below:

		X			
		-1	0	1	
Y	-1	0.09	0.12	0.09	0.30
	0	0.12	0.16	0.12	0.40
	1	0.09	0.12	0.09	0.30
		0.30	0.40	0.30	1.00

- iv) For X and Y to be independent, $P(X = i, Y = j) = P(X = i) * P(Y = j)$ for all i, j .

$$P(X = -1, Y = -1) = 0.09 = 0.30 * 0.30 = P(X = -1) * P(Y = -1)$$

$$P(X = -1, Y = 0) = 0.12 = 0.30 * 0.40 = P(X = -1) * P(Y = 0)$$

$$P(X = -1, Y = 1) = 0.09 = 0.30 * 0.30 = P(X = -1) * P(Y = 1)$$

$$P(X = 0, Y = -1) = 0.12 = 0.40 * 0.30 = P(X = 0) * P(Y = -1)$$

$$P(X = 0, Y = 0) = 0.16 = 0.40 * 0.40 = P(X = 0) * P(Y = 0)$$

$$\begin{aligned}
P(X = 0, Y = 1) &= 0.12 = 0.40 * 0.30 = P(X = 0) * P(Y = 1) \\
P(X = 1, Y = -1) &= 0.09 = 0.30 * 0.30 = P(X = 1) * P(Y = -1) \\
P(X = 1, Y = 0) &= 0.12 = 0.30 * 0.40 = P(X = 1) * P(Y = 0) \\
P(X = 1, Y = 1) &= 0.09 = 0.30 * 0.30 = P(X = 1) * P(Y = 1)
\end{aligned}$$

Thus, X and Y are independent.

[12]

Q. 6) i) X represents the claim amount which follows an Exponential distribution with mean θ . This means:

- $E(X) = \theta$
- $Var(X) = \theta^2$

q is the probability of a claim for an insured population of size w and N represents the number of claims. This means:

- N is a Binomial random variable with parameters w and q .
- $E(N) = wq$
- $Var(N) = wq(1 - q)$

S is the total reported claims. Using the fact that X and N are independent random variables, S is the compound distribution, with:

- $E(S) = E(N)E(X) = wq\theta$
- $Var(S) = E(N)Var(X) + Var(N)(E(X))^2 = wq\theta^2 + wq(1 - q)\theta^2 = wq(2 - q)\theta^2$

ii) The summary statistics is given as below:

Age Group	Proportion	Death Count ¹	Average Benefit ²
18 - 35	50%	25	5.0
36 - 50	30%	75	6.0
51 - 65	20%	100	7.5

¹ **Death Count:** Average number of deaths per 1000 lives over the last 10 years

² **Average Benefit:** Average claim amount (in Lakhs) paid on death of an employee over the last 10 years

As the actuary believes that the claim sizes usually follow an Exponential distribution, the average benefit amount would be an estimate for the mean θ for the Exponential random variables (X_i) representing each age group.

For the age-group i , the probability of claim (q_i) can be estimated by Death Count/1000. The random variables representing the number of claims in age-group will be a Binomial random variable with parameters $w_i = 1000 * prop_i$ and q_i . Here: $prop_i$ represents the proportion of the population of 1000 employees within age-group i .

Denote S as the total claims for the insurer.

Then, $S = S_1 + S_2 + S_3$ where S_1 , S_2 and S_3 are the total claims reported per person for age-groups 18 – 35, 36 – 50 and 51 – 65 respectively.

Since the age-groups are independent,

- $E(S) = E(S_1) + E(S_2) + E(S_3)$
- $Var(S) = Var(S_1) + Var(S_2) + Var(S_3)$

Using the data above and the formulas obtained in part (i), and using the fact that all lives, the number of claims and size of claims are independent across all age groups, we get:

Age Group	w_i	q_i	θ_i	$E(S_i)$	$Var(S_i)$
18 - 35	500	0.025	5.0	62.50	617.19
36 - 50	300	0.075	6.0	135.00	1,559.25
51 - 65	200	0.100	7.5	150.00	2,137.50
				347.50	4,313.94

Thus:

- $E(S) = 347.50$
- $Var(S) = 4313.94$

Using normal approximation, $S \sim N(347.50, 4313.94)$

We need the premium P to be set such that:

$$\begin{aligned}
 0.05 &= P(S > P) \\
 &= P\left(\frac{S - 347.50}{\sqrt{4313.94}} > \frac{P - 347.50}{\sqrt{4313.94}}\right) \\
 &= P\left(Z > \frac{P - 347.50}{\sqrt{4313.94}}\right) \\
 &= 1 - \Phi\left(\frac{P - 347.50}{\sqrt{4313.94}}\right) \quad \dots Z \sim N(0,1)
 \end{aligned}$$

Using normal probability tables,

$$\frac{P - 347.50}{\sqrt{4313.94}} = 1.6449$$

$$\text{i.e., } P = 347.50 + 1.6449 * \sqrt{4313.94} = 455.53 \text{ Lakhs.}$$

[7]

Q. 7) $X_1, X_2 \dots X_{2n}$ are independent observations from the Bernoulli distribution with unknown parameter p ($0 < p < 1$).

The estimator considered is:

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n Z_i \quad \text{where } Z_i = X_{2i-1} X_{2i} \text{ for } i = 1, 2 \dots n.$$

i) The values Z_i can take are:

$$Z_i = X_{2i-1} X_{2i} = \begin{cases} 1 & \text{if } X_{2i-1} = X_{2i} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} P(Z_i = 1) &= P(X_{2i-1} = 1, X_{2i} = 1) \\ &= P(X_{2i-1} = 1) * P(X_{2i} = 1) \quad \dots \text{ using independence property of } X_{j-s} \\ &= p^2 \\ &= \lambda \end{aligned}$$

$$\begin{aligned} P(Z_i = 0) &= 1 - P(Z_i = 1) \\ &= 1 - \lambda \end{aligned}$$

For every pair (Z_i, Z_j) $i \neq j$, Z_i can be considered independent of Z_j as (X_{2i-1}, X_{2i}) are independent of (X_{2j-1}, X_{2j}) .

Thus, $Z_1, Z_2 \dots Z_n$ can be considered as independent observations from the Bernoulli distribution with parameter λ .

ii)

$$\begin{aligned} E(\hat{\lambda}) &= E\left(\frac{1}{n} \sum_{i=1}^n Z_i\right) \\ &= \frac{1}{n} \sum_{i=1}^n E(Z_i) \\ &= \lambda \quad \dots \text{ using the fact that expectation of a Bernoulli random variable } Z_i \text{ is } \lambda \end{aligned}$$

Thus, $\hat{\lambda}$ is an unbiased estimator of λ .

iii) We can use the random sample $Z_1, Z_2 \dots Z_n$ to construct the likelihood function of λ .

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^n \lambda^{Z_i} (1 - \lambda)^{1-Z_i} \\ &= \lambda^{\sum Z_i} (1 - \lambda)^{n - \sum Z_i} \end{aligned}$$

Taking logarithms, the log-likelihood function is given by:

$$\log_e L(\lambda) = \left(\sum_{i=1}^n Z_i\right) \log_e \lambda + \left(n - \sum_{i=1}^n Z_i\right) \log_e (1 - \lambda)$$

Differentiating w.r.t. λ ,

$$\frac{d}{d\lambda} \log_e L(\lambda) = \left(\sum_{i=1}^n Z_i\right) \left(\frac{1}{\lambda}\right) + \left(n - \sum_{i=1}^n Z_i\right) \left(\frac{-1}{1 - \lambda}\right)$$

Differentiating w.r.t. λ again,

$$\frac{d^2}{d\lambda^2} \log_e L(\lambda) = \left(\sum_{i=1}^n Z_i\right) \left(\frac{-1}{\lambda^2}\right) + \left(n - \sum_{i=1}^n Z_i\right) \left[\frac{(-1)(-1)(-1)}{(1 - \lambda)^2}\right]$$

Computing expectation for this,

$$\begin{aligned}
 E \left[\frac{d^2}{d\lambda^2} \log_e L(\lambda) \right] &= -\frac{1}{\lambda^2} \sum_{i=1}^n E(Z_i) - \frac{1}{(1-\lambda)^2} \left[n - \sum_{i=1}^n E(Z_i) \right] \\
 &= -\frac{1}{\lambda^2} \sum_{i=1}^n \lambda - \frac{1}{(1-\lambda)^2} \left[n - \sum_{i=1}^n \lambda \right] \\
 &= -\frac{n}{\lambda} - \frac{n}{1-\lambda} \\
 &= -\frac{n}{\lambda(1-\lambda)}
 \end{aligned}$$

Thus, the Cramér-Rao lower bound for the variance of unbiased estimators of λ is given by:

$$\begin{aligned}
 CRLb &= -\frac{1}{E \left[\frac{d^2}{d\lambda^2} \log_e L(\lambda) \right]} \\
 &= \frac{\lambda(1-\lambda)}{n}.
 \end{aligned}$$

iv) $\hat{\lambda}$ is an unbiased estimator of λ with variance:

$$\begin{aligned}
 Var(\hat{\lambda}) &= Var\left(\frac{1}{n} \sum_{i=1}^n Z_i\right) \\
 &= \frac{1}{n^2} \sum_{i=1}^n Var(Z_i) \quad \dots \text{using the fact that } Z_i\text{'s are independent.} \\
 &= \frac{1}{n^2} \sum_{i=1}^n \lambda(1-\lambda) \\
 &= \frac{\lambda(1-\lambda)}{n}
 \end{aligned}$$

Thus we see that the variance of $\hat{\lambda}$ attains the Cramér-Rao lower bound.

[10]

Q. 8) The random variable $X \sim N(\theta, \theta^2)$.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{25} x_i = \frac{69.66}{25} = 2.79$$

The pivotal quantity is:

$$T = \frac{\sqrt{n}(\bar{X} - \theta)}{\theta} \sim N(0, 1)$$

Alternately, if one uses:

- t-pivot $\frac{\sqrt{n}(\bar{X} - \theta)}{S} \sim t_{24}$, CI: (2.13, 3.45)
- χ^2 -pivot $\frac{(n-1)S^2}{\theta^2} \sim \chi_{24}^2$, CI: (1.25, 2.23)

To obtain a 95% confidence interval for θ we note that:

$$P(-1.96 < T < 1.96) = 0.95$$

$$i.e. \quad P\left(-1.96 < \frac{\sqrt{n}(\bar{X} - \theta)}{\theta} < 1.96\right) = 0.95$$

$$i.e. P\left(\theta - \frac{1.96\theta}{\sqrt{n}} < \bar{X} < \theta + \frac{1.96\theta}{\sqrt{n}}\right) = 0.95$$

Therefore, a 95% confidence interval for θ is given by:

$$\frac{2.79}{1 + \frac{1.96}{\sqrt{25}}} < \theta < \frac{2.79}{1 - \frac{1.96}{\sqrt{25}}} \quad i.e. \quad 2.00 < \theta < 4.59$$

[5]

Q. 9) i) $\sum x = 1,239 \Rightarrow \bar{x} = 154.875$
 $\sum y = 1,105 \Rightarrow \bar{y} = 138.125$

$$\sum x^2 = 194,095; \quad \sum \sum xy = 172,880$$

$$S_{xx} = \sum x^2 - \left(\sum x\right)^2 / 8 = 194095 - 1239^2 / 8 = 2204.875$$

$$S_{xy} = \sum \sum xy - \left(\sum x\right)\left(\sum y\right) / 8 = 172880 - 1239 * 1105 / 8 = 1743.125$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{1743.125}{2204.875} = 0.7906$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} * \bar{x} = 138.125 - 0.7906 * 154.875 = 15.684$$

Hence, the fitted regression equation of y on x is:

$$Y = 15.684 + 0.7906 x$$

ii) $\sum y^2 = 154,141$
 $S_{yy} = \sum y^2 - \left(\sum y\right)^2 / 8 = 154141 - 1105^2 / 8 = 1512.875$

Thus, an estimate of the error variance σ^2 is given by:

$$\hat{\sigma}^2 = \frac{1}{n-2} \left(S_{yy} - \frac{S_{xy}^2}{S_{xx}} \right) = \frac{1}{6} * \left(1512.875 - \frac{1743.125^2}{2204.875} \right) = 22.4665$$

Assuming the full Normal model,

$$\frac{6 \hat{\sigma}^2}{\sigma^2} \sim \chi^2 \text{ distribution with d.f. } 6$$

To obtain a 90% confidence interval for σ^2 , we note:

$$P(1.635 < \chi_6^2 < 12.59) = 0.90$$

$$P\left(1.635 < \frac{6 \hat{\sigma}^2}{\sigma^2} < 12.59\right) = 0.90$$

Thus, the 90% confidence interval for σ^2 is given by:

$$\left(\frac{6 * 22.4665}{12.59} \quad \frac{6 * 22.4665}{1.635} \right)$$

i.e., (10.7068 82.4459)

- iii) The proportion of variation explained by the model is given by the coefficient of determination, denoted by R^2 :

$$R^2 = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \frac{1743.125^2}{2204.875 * 1512.875} = 0.9109$$

Comment:

91.09% of the variation is explained by the model, which indicates that the fit is quite good. It still might be worthwhile to examine the residuals to double check that a linear model is appropriate.

[10]

- Q.10)** i) The estimate of the overall mean temperature is given by:

$$\begin{aligned} \bar{y}_{..} &= \frac{\sum \sum y_{ij}}{(n_1 + n_2 + n_3)} \\ &= \frac{(395 + 473 + 365)}{(8 + 6 + 7)} \\ &= \frac{1233}{21} \\ &= 58.7^\circ\text{C} \end{aligned}$$

- ii) We are testing the hypotheses:

H_0 : Mean temperature is same for each of the three irons

v/s

H_1 : There are differences between the mean levels of temperatures between the three irons

$$SSR = SST - SSB$$

$$\begin{aligned} SST &= \sum \sum y_{ij}^2 - \left(\sum \sum y_{ij} \right)^2 / (n_1 + n_2 + n_3) \\ &= (19975 + 37435 + 19225) - \frac{1233^2}{21} \\ &= 4240.29 \end{aligned}$$

$$\begin{aligned} SSB &= \sum_i \left[\left(\sum_j y_{ij} \right)^2 / n_i \right] - \left(\sum \sum y_{ij} \right)^2 / (n_1 + n_2 + n_3) \\ &= \left(\frac{395^2}{8} + \frac{473^2}{6} + \frac{365^2}{7} \right) - \frac{1233^2}{21} \\ &= 3428.72 \end{aligned}$$

$$\begin{aligned} SSR &= SST - SSB \\ &= 4240.29 - 3428.72 \\ &= 811.57 \end{aligned}$$

Thus, the ANOVA table is as below:

ANOVA				
<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between Groups	3,428.72	2	1,714.36	38.02
Within Groups	811.57	18	45.09	
Total	4,240.29	20		

The MS ratio is 38.02, which has a $F_{2,18}$ distribution

5% critical value for $F_{2,18}$ is 3.555, which is much less than 38.02.
Hence, we have sufficient evidence to reject H_0 at the 5% level.

Assumptions made:

- Underlying distribution is Normal
- Population have common variance
- Observations are independent.

We thereby conclude that there are the differences among the three means cannot be attributed to chance.

iii) An estimate of the underlying common variance in temperature readings is given by:

$$\begin{aligned}
 \hat{\sigma}^2 &= \left(\frac{1}{n-k} \right) * SS_R \\
 &= \frac{1}{18} * 811.57 \\
 &= 45.09.
 \end{aligned}$$

[8]

Q. 11) i) The observed proportions are $x_1, x_2 \dots x_{100}$ which are random observations from the following PDF:

$$f(x; \theta) = \theta(\theta + 1)x^{\theta-1}(1 - x); \quad 0 < x < 1 \quad \text{and} \quad \theta > 0 \text{ is unknown}$$

The likelihood function of θ is given as below:

$$\begin{aligned}
 L(\theta) &= \prod_{i=1}^{100} f(x_i; \theta) \\
 &= \prod_{i=1}^{100} \theta(\theta + 1)(x_i)^{\theta-1}(1 - x_i) \\
 &= \theta^{100}(\theta + 1)^{100} \prod_{i=1}^{100} (x_i)^{\theta-1}(1 - x_i)
 \end{aligned}$$

Taking logarithms, the log-likelihood function is given by:

$$\log_e L(\theta) = 100 \log_e \theta + 100 \log_e(\theta + 1) + (\theta - 1) \sum_{i=1}^{100} \log_e x_i + \sum_{i=1}^{100} \log_e(1 - x_i)$$

Differentiating w.r.t. θ ,

$$\frac{d}{d\theta} \log_e L(\theta) = \frac{100}{\theta} + \frac{100}{\theta + 1} + \sum_{i=1}^{100} \log_e x_i$$

To obtain the MLE, we must have:

$$\frac{d}{d\theta} \log_e L(\theta) = 0$$

$$i.e., \quad \frac{100}{\theta} + \frac{100}{\theta + 1} + \sum_{i=1}^{100} \log_e x_i = 0$$

$$i.e., \quad \frac{2\theta + 1}{\theta(\theta + 1)} + \frac{1}{100} \sum_{i=1}^{100} \log_e x_i = 0$$

$$i.e., \quad \theta^2 + (2k + 1)\theta + k = 0 \quad \text{where} \quad k^{-1} = \frac{1}{100} \sum_{i=1}^{100} \log_e x_i$$

Thus, $\hat{\theta}$ satisfies the given quadratic equation.

$$ii) \quad \sum_{i=1}^{100} \log_e x_i = -58.334 \Rightarrow k = -\frac{100}{58.334} = -1.714.$$

The MLE of θ is given by one of the roots of the quadratic equation:

$$\theta = \frac{-(2k + 1) \pm \sqrt{(2k + 1)^2 - 4k}}{2} = \frac{2.428 \pm \sqrt{12.75118}}{2} = (-0.571, 3.000)$$

Since $\theta > 0$, the MLE of θ is 3.

Check: Differentiating w.r.t. θ again,

$$\frac{d^2}{d\theta^2} \log_e L(\theta) = -\frac{100}{\theta^2} - \frac{100}{(\theta + 1)^2} < 0 \quad \text{for all values of } \theta$$

iii) Using the MLE of θ , we can estimate the value of $\phi_1(\theta)$ as:

$$\begin{aligned} \hat{\phi}_1(\theta) &= P(X \leq 0.25 \mid \theta = \hat{\theta}) \\ &= \int_0^{0.25} \hat{\theta}(\hat{\theta} + 1)x^{\hat{\theta}-1}(1-x) dx \\ &= \int_0^{0.25} 12x^2(1-x) dx \\ &= [4x^3 - 3x^4]_0^{0.25} \\ &= 0.063 - 0.012 \\ &= 0.051. \end{aligned}$$

$$\begin{aligned}
 \hat{\phi}_2 &= 1 - \hat{\phi}_1 - \sum_{j=3}^7 \hat{\phi}_j \\
 &= 1 - 0.051 - (0.163 + 0.176 + 0.167 + 0.128 + 0.052) \\
 &= 0.263
 \end{aligned}$$

- iv) If the proportion of defective items follow the distribution with the given PDF $f(x; \theta)$, the estimated expected frequencies ($= 100 * \hat{\phi}_j$) for the 7 categories are:

Category	Observed Frequency	Expected Frequency
0.00 - 0.25	9	5.1
0.25 - 0.50	20	26.3
0.50 - 0.60	14	16.3
0.60 - 0.70	21	17.6
0.70 - 0.80	17	16.7
0.80 - 0.90	14	12.8
0.90 - 1.00	5	5.2

We conduct a χ^2 _Goodness-of-Fit test using the following test statistic:

$$\chi^2 = \sum_{i=1}^7 \frac{(O_i - E_i)^2}{E_i} \sim \text{chi_square distribution with d.f. } 5 (= 7 - 1 - 1)$$

Computations are as below:

Category	Observed Frequency	Expected Frequency	O - E	χ^2
0.00 - 0.25	9	5.1	3.9	2.982
0.25 - 0.50	20	26.3	-6.3	1.509
0.50 - 0.60	14	16.3	-2.3	0.325
0.60 - 0.70	21	17.6	3.4	0.657
0.70 - 0.80	17	16.7	0.3	0.005
0.80 - 0.90	14	12.8	1.2	0.113
0.90 - 1.00	5	5.2	-0.2	0.008
				5.599

Since the value of the observed test statistic is less than the critical value of $\chi^2(5; 0.05) = 11.07$, we conclude that there is no evidence at the 5% level of significance that the data do not conform to the assumed model.

[16]

- Q. 12) i) The k^{th} -measurement of weight by the first electronic scale are given by ($k = 1, 2 \dots 10$): $X_k = 100 + F_k^X$, where $F_k^X \sim N(0, \sigma^2)$

This is equivalent to:

$$X_k \sim N(100, \sigma^2)$$

Thus, the distribution of $\left(\frac{X_k - 100}{\sigma}\right) \sim N(0, 1)$

$$\Rightarrow \left(\frac{X_k - 100}{\sigma}\right)^2 \sim \chi_1^2$$

Given $X_1, X_2 \dots X_{10}$ are independent observations,

$$V = \sum_{i=1}^{10} \left(\frac{X_k - 100}{\sigma}\right)^2 \sim \chi_{10}^2$$

Similarly, we get:

$$Y_k = 100 + F_k^Y, \text{ where } F_k^Y \sim N(0, \psi^2) \text{ for } k = 1, 2 \dots 8$$

$$\Rightarrow \left(\frac{Y_k - 100}{\psi}\right) \sim N(0, 1)$$

$$\Rightarrow \left(\frac{Y_k - 100}{\psi}\right)^2 \sim \chi_1^2$$

Given $Y_1, Y_2 \dots Y_8$ are independent observations,

$$W = \sum_{i=1}^8 \left(\frac{Y_k - 100}{\psi}\right)^2 \sim \chi_8^2$$

ii)
$$U = \frac{\sigma^2 (V/10)}{\psi^2 (W/8)}$$

V and W can be considered as independent random variables as the measurements were taken in two different electronic scales.

By definition of a F-distribution,

$$\frac{(V/10)}{(W/8)} \sim F_{10,8} \Leftrightarrow U \frac{\psi^2}{\sigma^2} \sim F_{10,8}$$

This can be regarded as the pivotal quantity for $\frac{\sigma^2}{\psi^2}$ because:

- It is a function of sample values (U) and unknown parameter $\theta = \frac{\sigma^2}{\psi^2}$
- It's distribution is completely known ($F_{10,8}$)
- It is monotonic in $\theta = \frac{\sigma^2}{\psi^2} \dots g(\theta) = \frac{U}{\theta}$

iii) To obtain a 95% confidence interval for $\frac{\sigma^2}{\psi^2}$, we note:

$$P(0.259 < F_{10,8} < 4.295) = 0.95$$

$$P\left(0.259 < U \frac{\psi^2}{\sigma^2} < 4.295\right) = 0.95$$

Now:

$$U = \frac{\sigma^2 (V/10)}{\psi^2 (W/8)} = \frac{\sum_{k=1}^{10} (X_k - 100)^2 / 10}{\sum_{k=1}^8 (Y_k - 100)^2 / 8} = \frac{64.16/10}{13.64/8} = 3.763$$

The computations are as below:

Obs #	X_k	$X_k - 100$	$(X_k - 100)^2$	Obs #	Y_k	$Y_k - 100$	$(Y_k - 100)^2$
1	100.6	0.6	0.36	1	99.4	-0.6	0.36
2	98.0	-2.0	4.00	2	102.3	2.3	5.29
3	101.2	1.2	1.44	3	100.7	0.7	0.49
4	102.0	2.0	4.00	4	98.8	-1.2	1.44
5	104.8	4.8	23.04	5	98.3	-1.7	2.89
6	99.4	-0.6	0.36	6	101.6	1.6	2.56
7	97.2	-2.8	7.84	7	99.5	-0.5	0.25
8	101.4	1.4	1.96	8	99.4	-0.6	0.36
9	95.4	-4.6	21.16				
10	100.0	0.0	0.00				
			64.16				13.64

Thus, the 95% confidence interval for $\frac{\sigma^2}{\psi^2}$ is given by:

$$\left(\frac{3.763}{4.295}, \frac{3.763}{0.259} \right) \quad \text{i.e., } (0.876, 14.529)$$

- iv) We need to conduct the test $H_0: \sigma^2 = \psi^2$ against $H_1: \sigma^2 \neq \psi^2$ at the 5% level.

The test statistic for this hypothesis test is: $T = U \frac{\psi^2}{\sigma^2}$

Under H_0 , $T = U \sim F_{10,8}$ distribution.

The observed value of T (under H_0) is 3.763 (from part iii).

The p-value for this hypothesis test can be derived by computing $\text{Prob}(F_{10,8} > 3.763)$.

$$\begin{aligned} P(F_{10,8} > 3.763) &= 1 - P(F_{10,8} \leq 3.763) \\ &\approx 1 - \left[0.96327 + \left(\frac{0.96934 - 0.96327}{4.00 - 3.75} \right) * (3.763 - 3.75) \right] \\ &= 1 - 0.96359 \\ &= 0.03641 \end{aligned}$$

As this test is two-sided, so the probability of obtaining a more extreme value than one actually obtained is $2 * 0.03641 = 0.07282$.

Thus, the p-value for this hypothesis test is 7.3% which is larger than 5% (significance level). Therefore, we can conclude that we do not have sufficient evidence to reject H_0 at the 5% level.

- v) From part (iii), the 95% confidence interval for $\frac{\sigma^2}{\psi^2}$ is (0.876, 14.529) which contains 1. This means one can conclude that the null hypothesis test $H_0: \sigma^2 = \psi^2$ can be accepted at the 5% level. This is consistent with the result obtained in part (iv).

[16]
